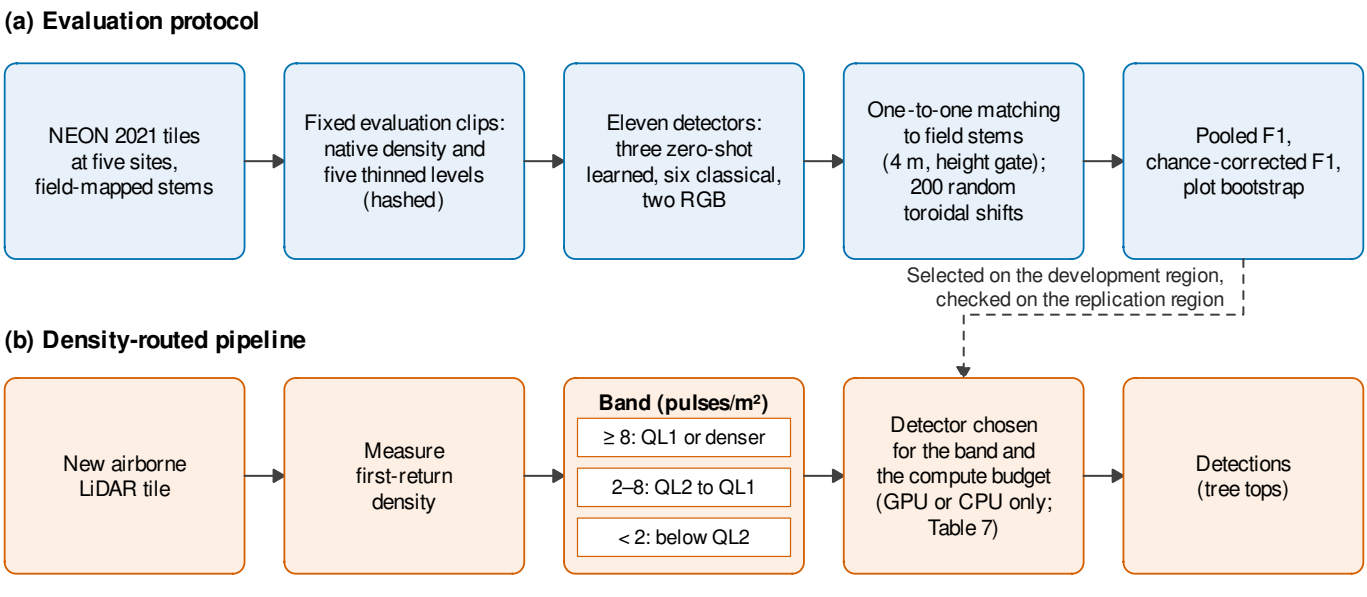
(a) Evaluation protocol
NEON 2021 tiles at five sites, field-mapped stems
Fixed evaluation clips: native density and five thinned levels (hashed)
Eleven detectors: three zero-shot learned, six classical, two RGB
One-to-one matching to field stems (4 m, height gate); 200 random toroidal shifts
Pooled F1, chance-corrected F1, plot bootstrap
Selected on the development region, checked on the replication region
(b) Density-routed pipeline
New airborne LiDAR tile
Measure first-return density
Band (pulses/m²)
≥ 8: QL1 or denser
2–8: QL2 to QL1
< 2: below QL2
Detector chosen for the band and the compute budget (GPU or CPU only; Table 7)
Detections (tree tops)